Eco-friendly gas test for optically readout TPCs

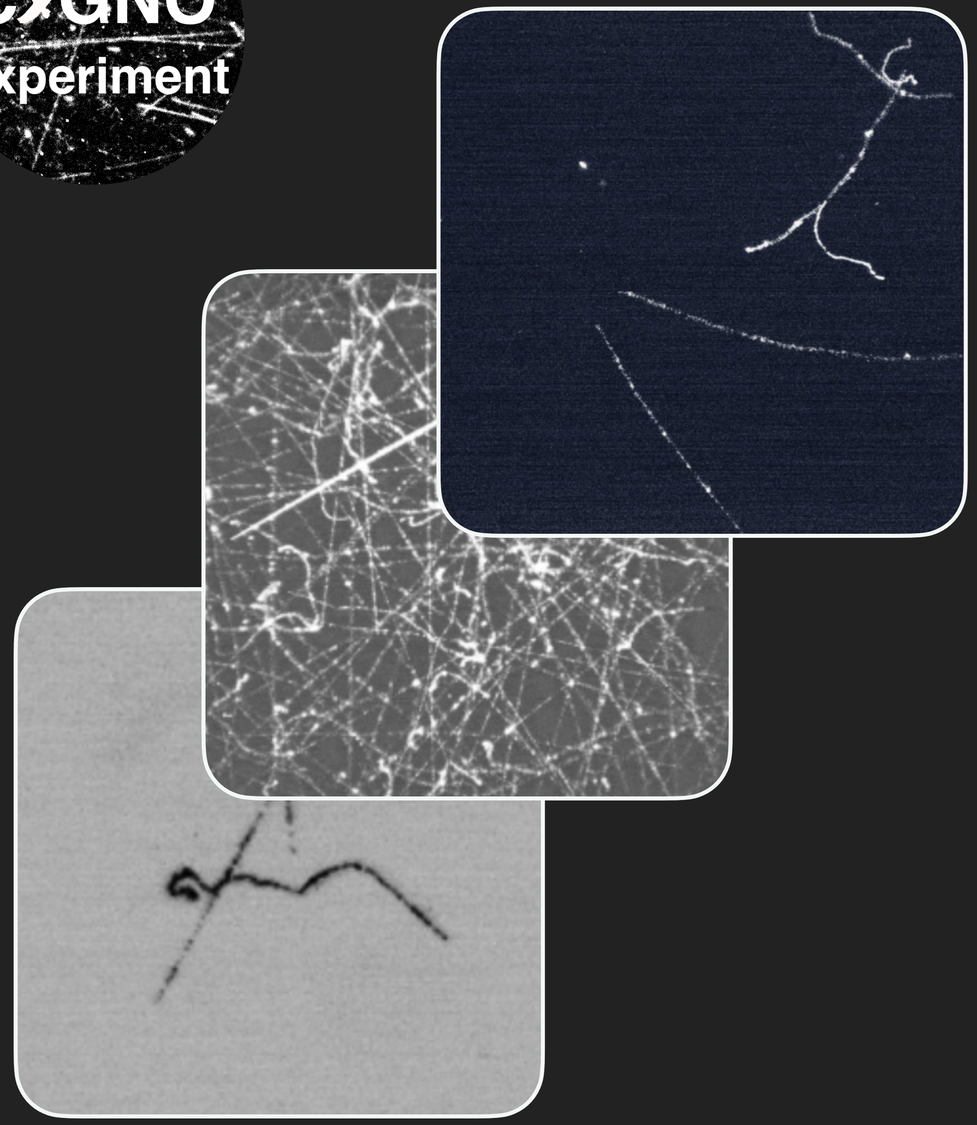
MELBA D'ASTOLFO

melba.dastolfo@gssi.it

GRAN SASSO SCIENCE INSTITUTE
SCHOOL OF ADVANCED STUDIES
Scuola Universitaria Superiore

INFN
LNGS

CYGNO Experiment



Gaseous TPC

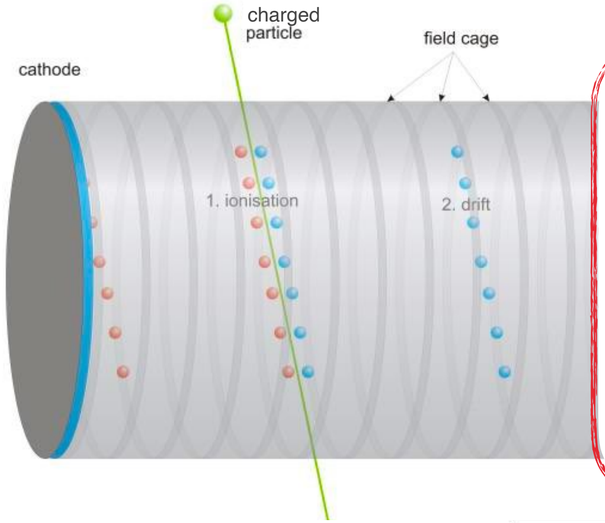
- He:CF_4 (60:40)
- Room temperature
- Atm pressure

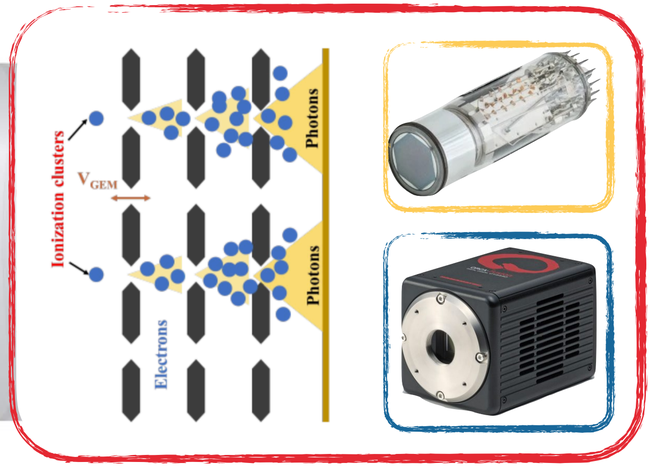
Triple GEM

Charge multiplication

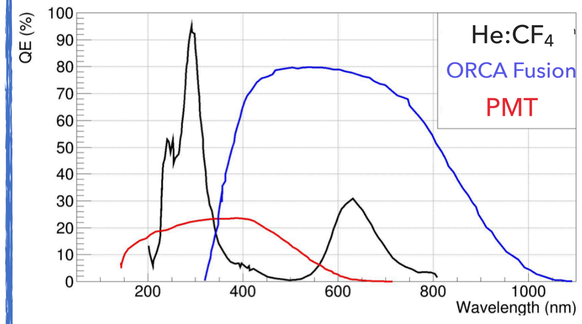
Camera + PMT

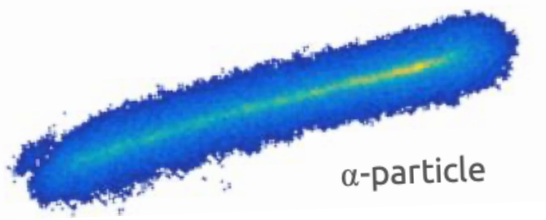
Light from gas scintillation during electron avalanche

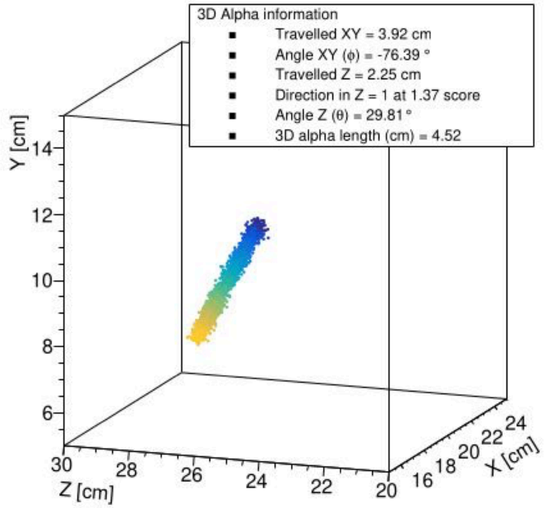




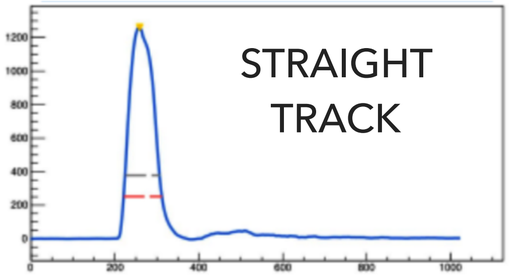
Energy + X & Y coordinates

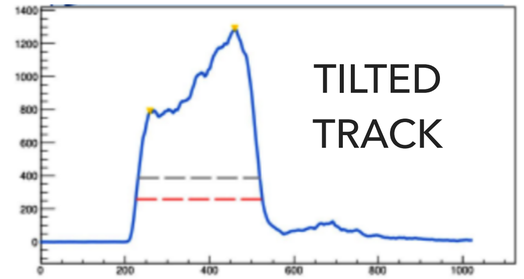




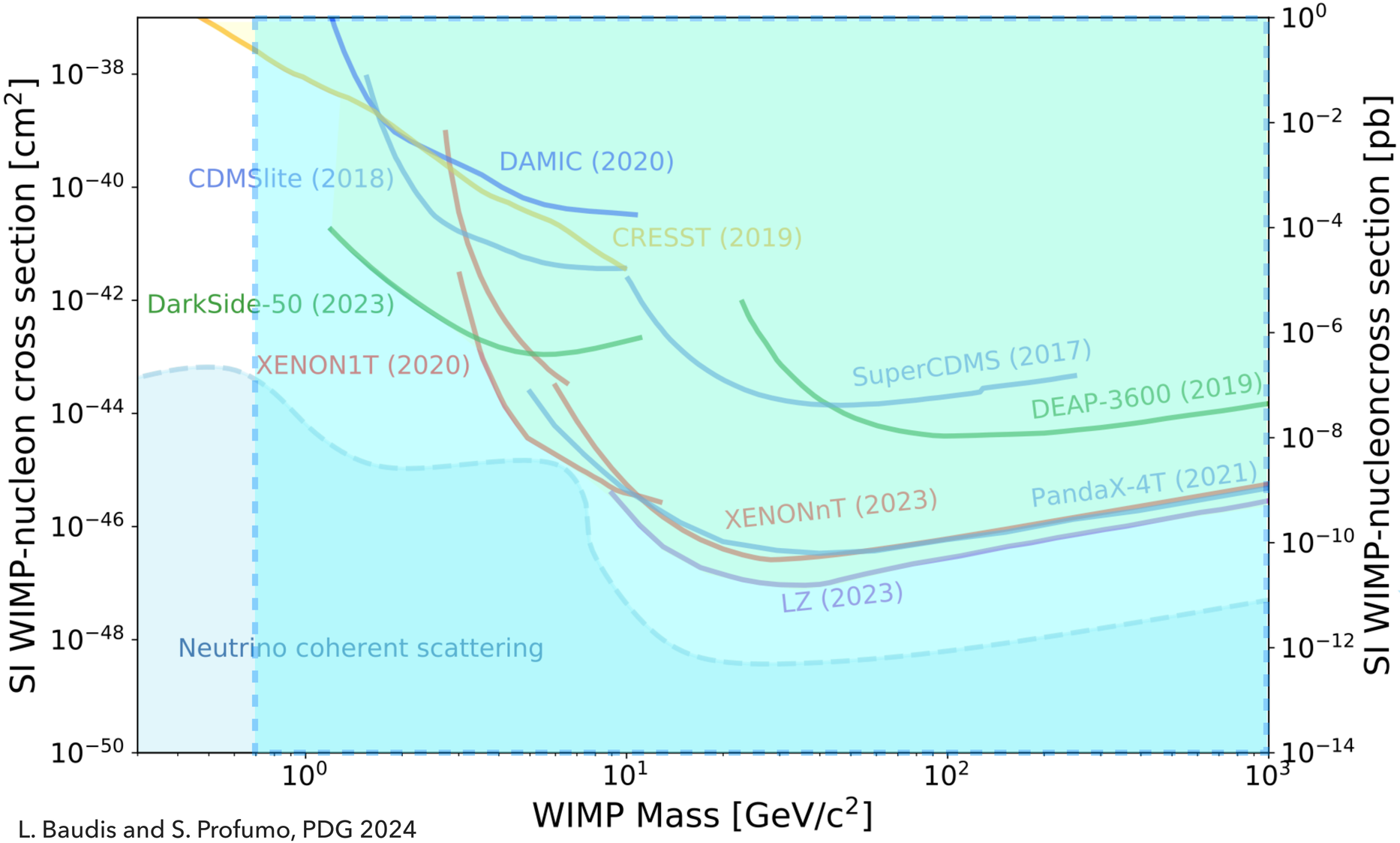


Integrated z + Energy





CYGNO Dark Matter exploration region



L. Baudis and S. Profumo, PDG 2024

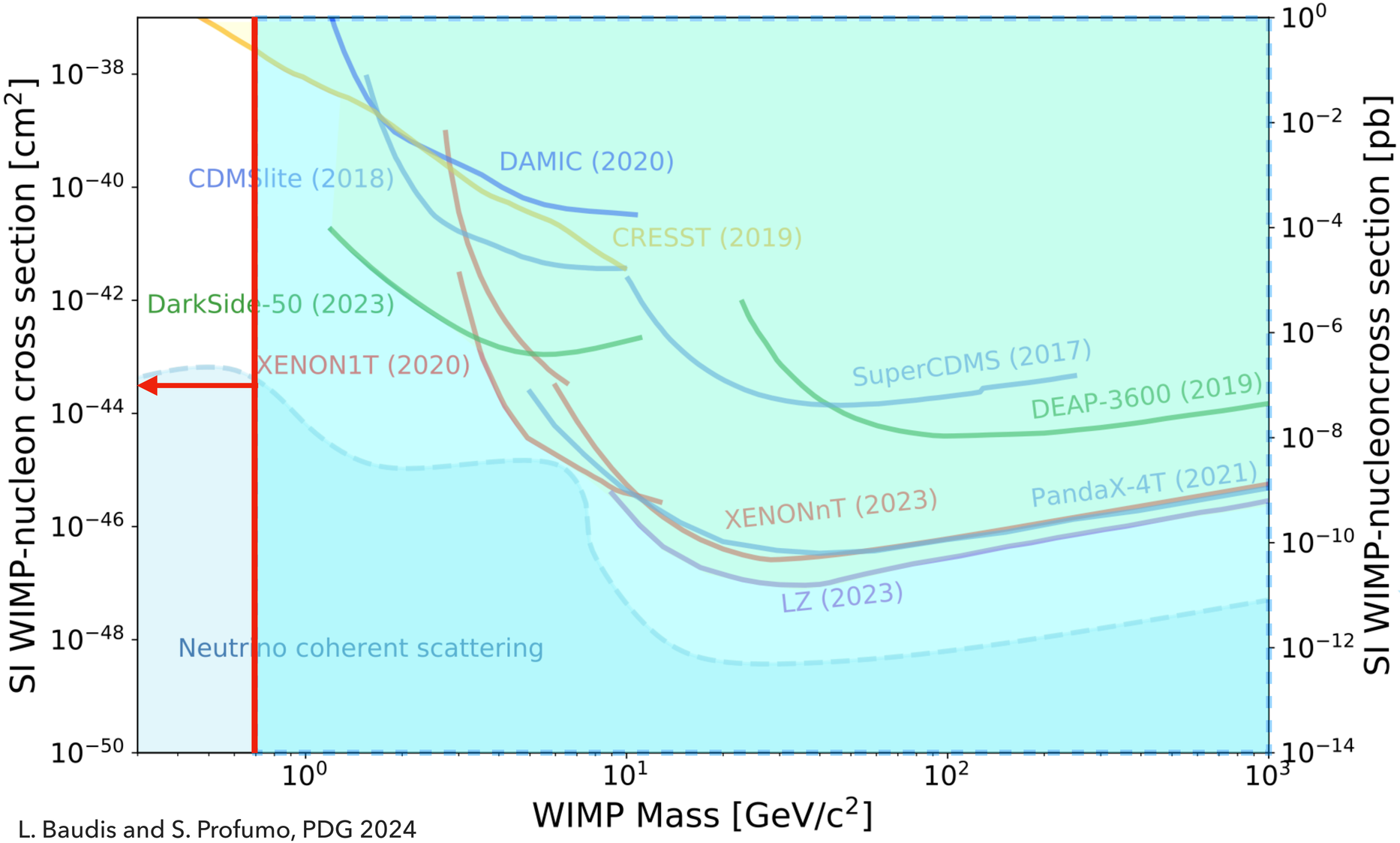
2
He

- ***Light*** target for low mass WIMPs
- Sensitive to ***SI*** couplings

9
F

- ***Heavier*** target for intermediate masses
- Sensitive to ***SD*** couplings

CYGNO Dark Matter exploration region



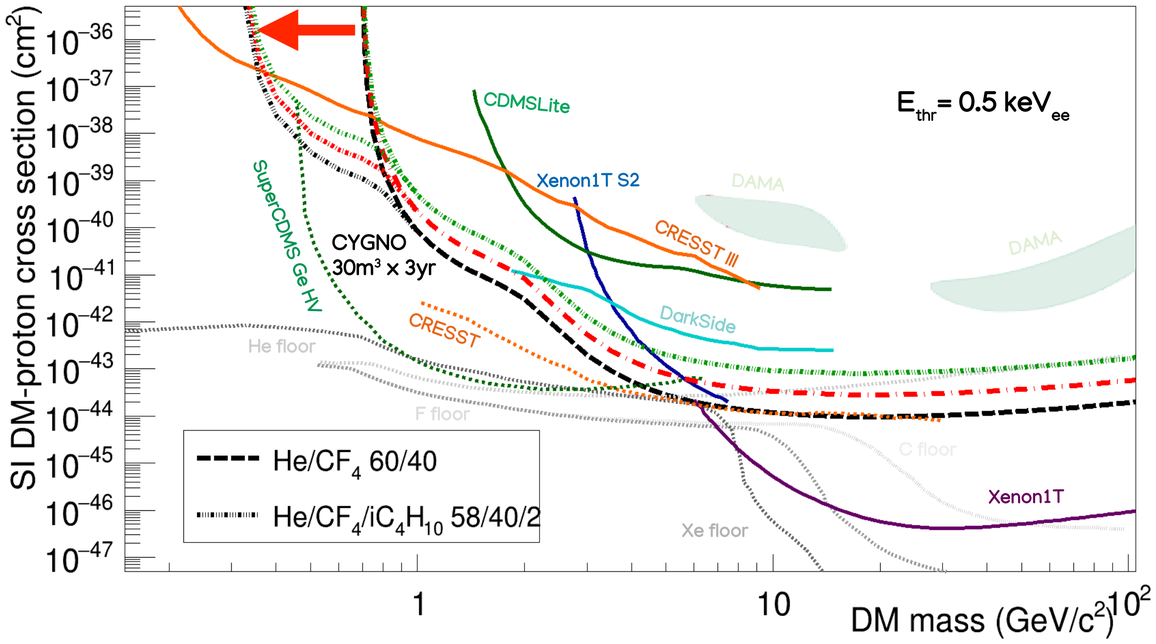
L. Baudis and S. Profumo, PDG 2024

2
He

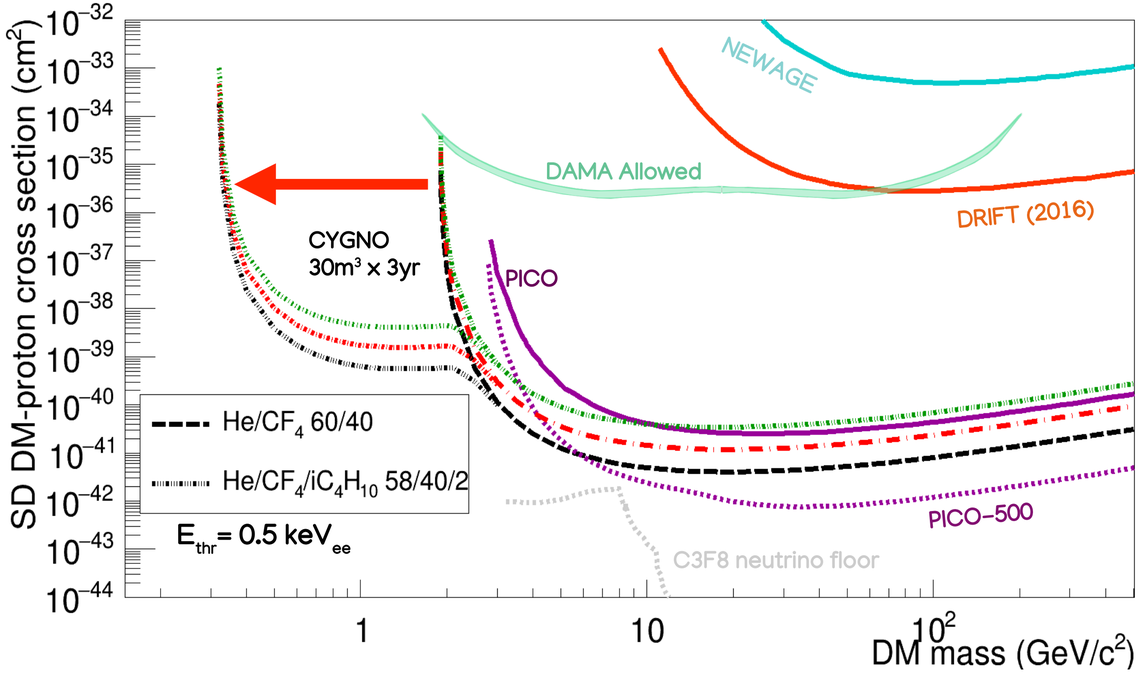
- ***Light*** target for low mass WIMPs
- Sensitive to ***SI*** couplings

9
F

- ***Heavier*** target for intermediate masses
- Sensitive to ***SD*** couplings



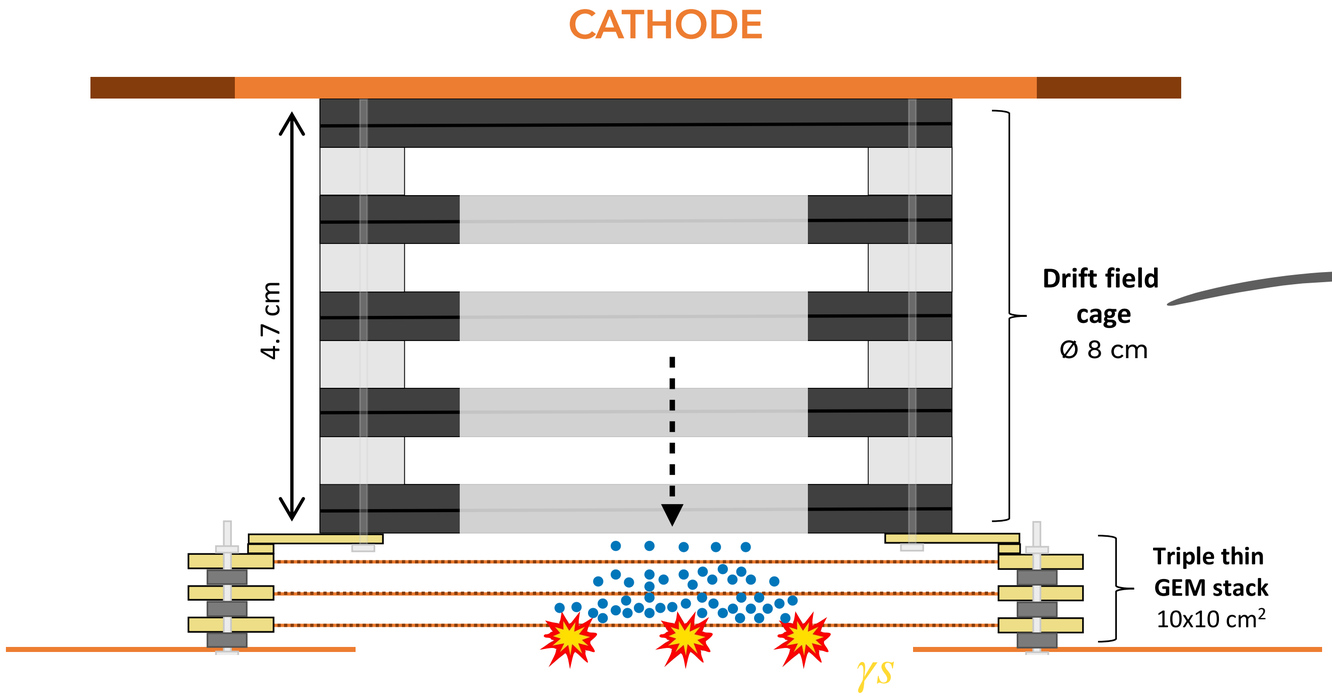
Consistent **improvement** both in **SI and SD exclusion** limits due to the presence of **Hydrogen**



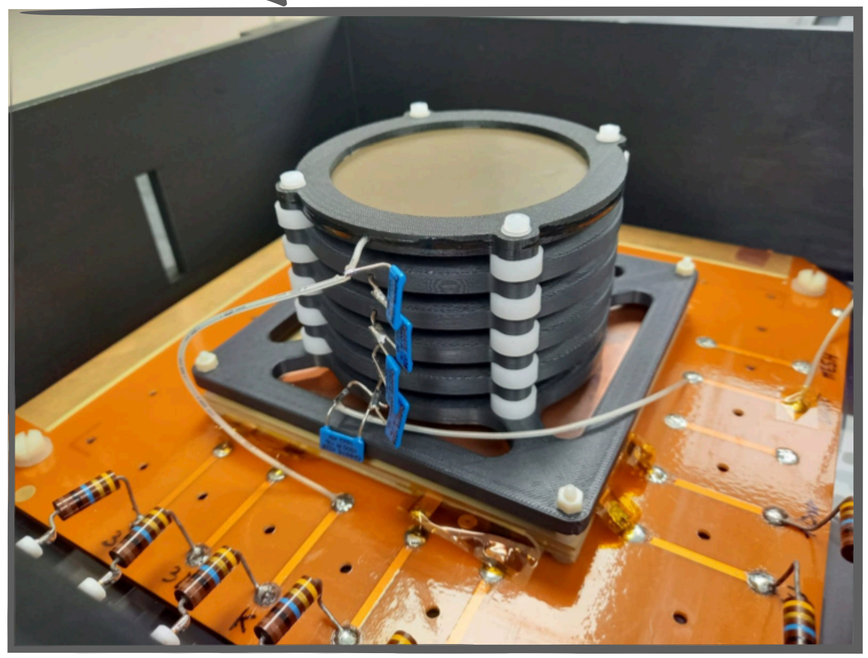
In our O(m³) demonstrator, CYGNO-04, the gas system is not suitable for hosting flammable gases ⇒ **HFO-1234ze**

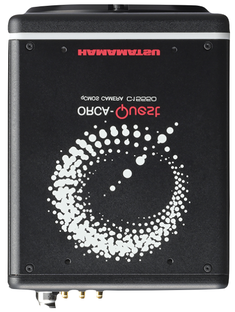
Secondary scintillation yield from GEM avalanches in He-CF4 and He-CF4-isobutane for CYGNO
CYGNO Collab. · Phys. Lett. B 858 (2024) 138759 · doi:10.1016/j.physletb.2024.138759

He-CF4-CH4 ternary mixtures as target gas for the CYGNO directional dark matter experiment
CYGNO Collab. · arXiv:2606.01512 (2026) · doi:10.48550/arXiv.2606.01512



MANGO: a Multipurpose Apparatus for Negative ion studies with GEM Optically readout

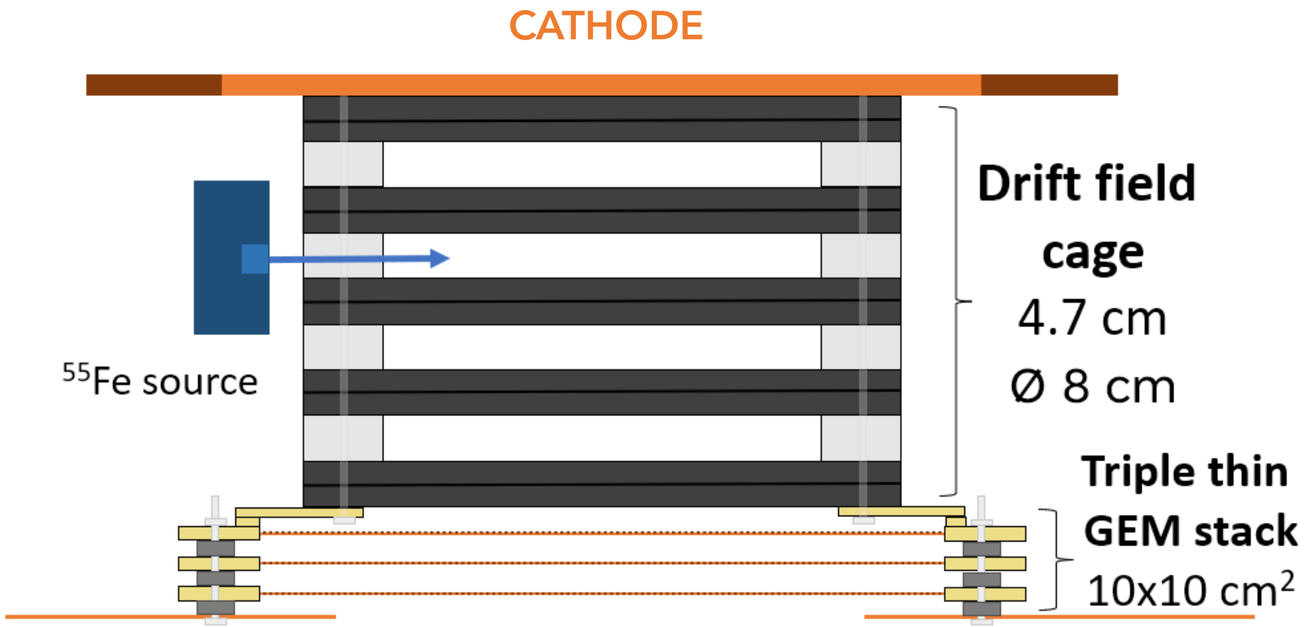




ORCA Quest



PMT



ELECTRON GAIN

$$G = \frac{I_{an}}{n_{Fe} e R}$$

I_{an} = Anode current

n_{Fe} = average pairs released by a 5.9 keV photon

e = elementary charge

R = interaction rate


ORCA Quest

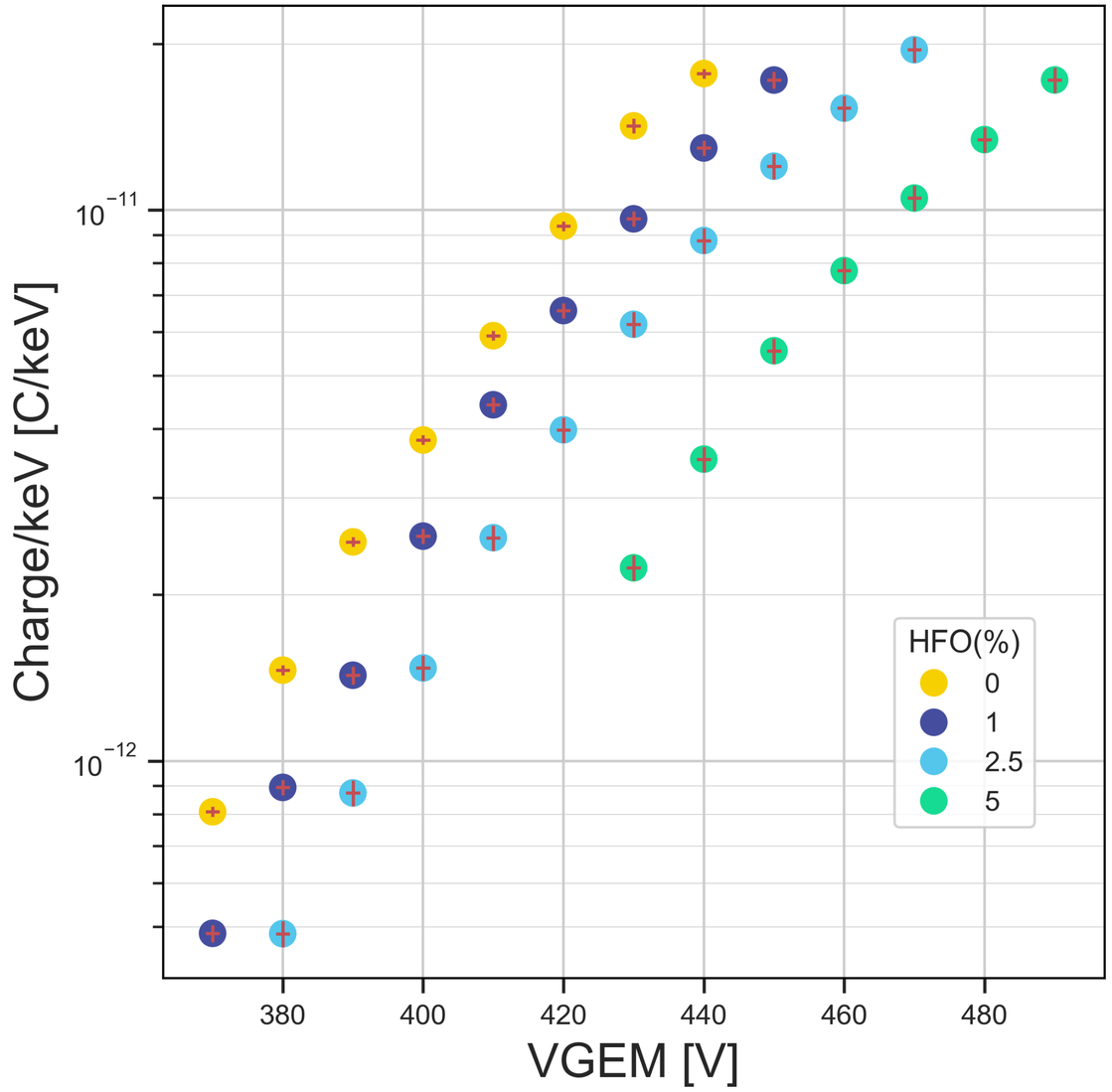

PMT

LIGHT YIELD

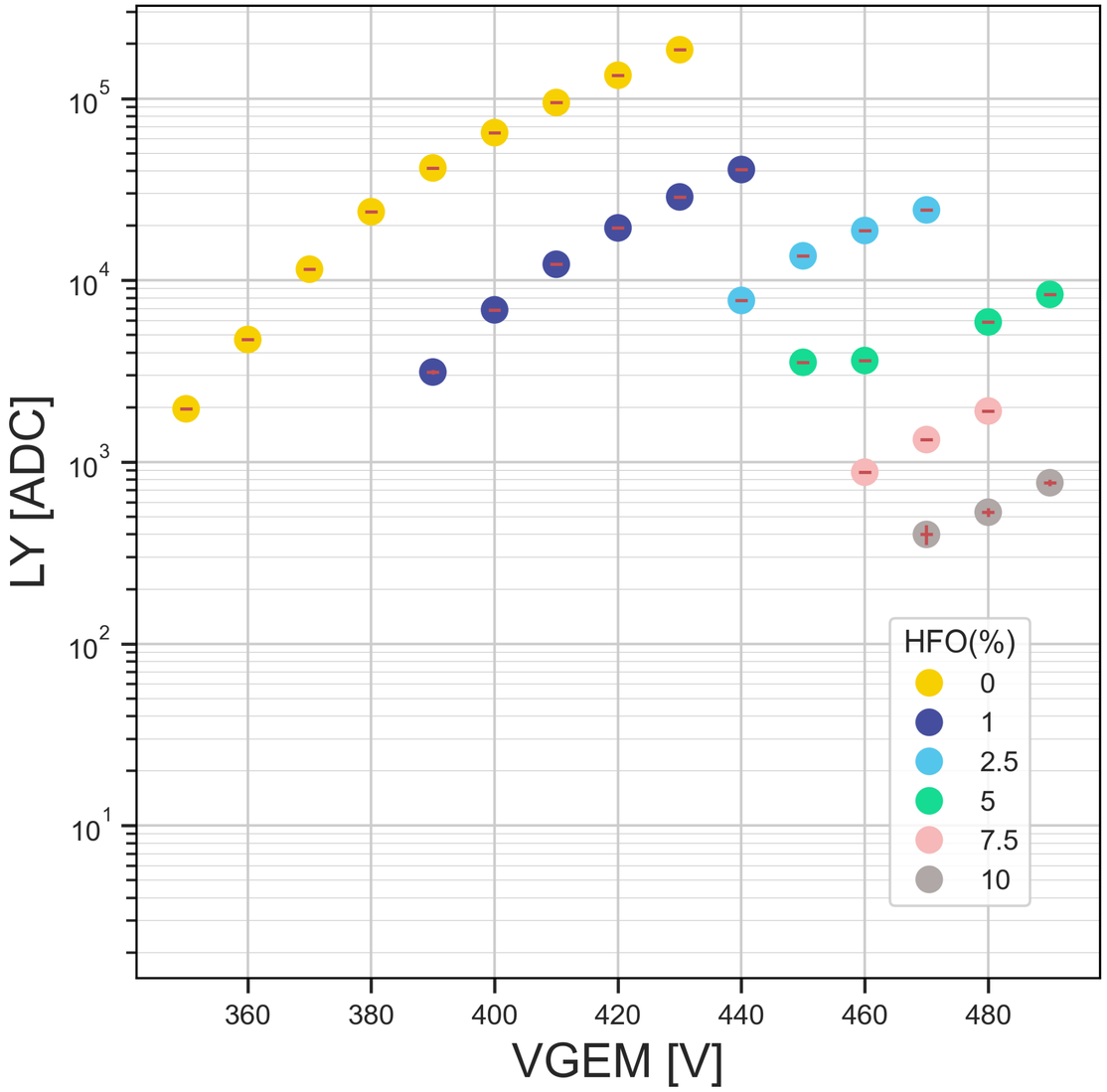
$$LY = avg(Integral)$$

$Integral$ = the sum of the intensities of all pixels belonging to a cluster of the reference iron spots

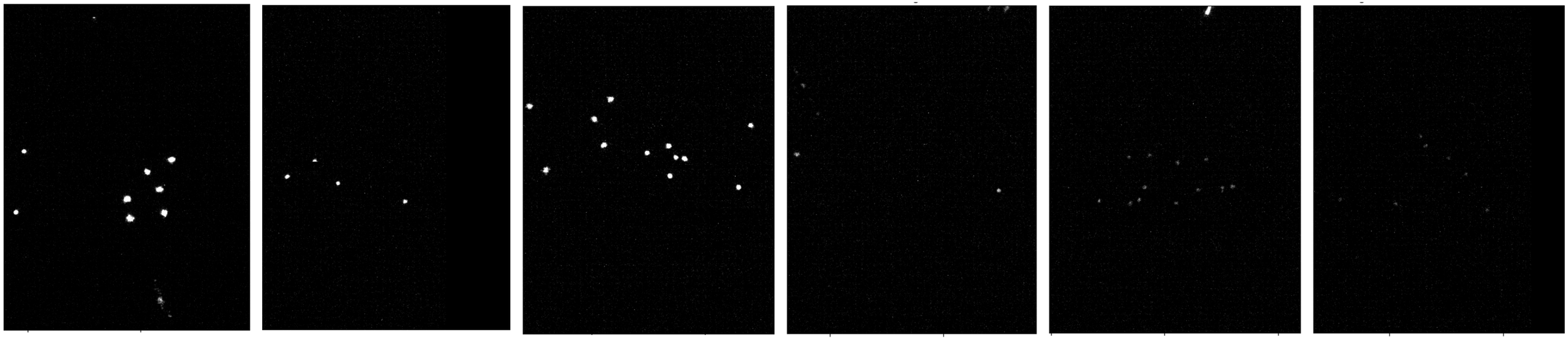
ELECTRON GAIN → ***recoverable*** at higher V_{GEM}



LIGHT YIELD → non recoverable → ***quenching***



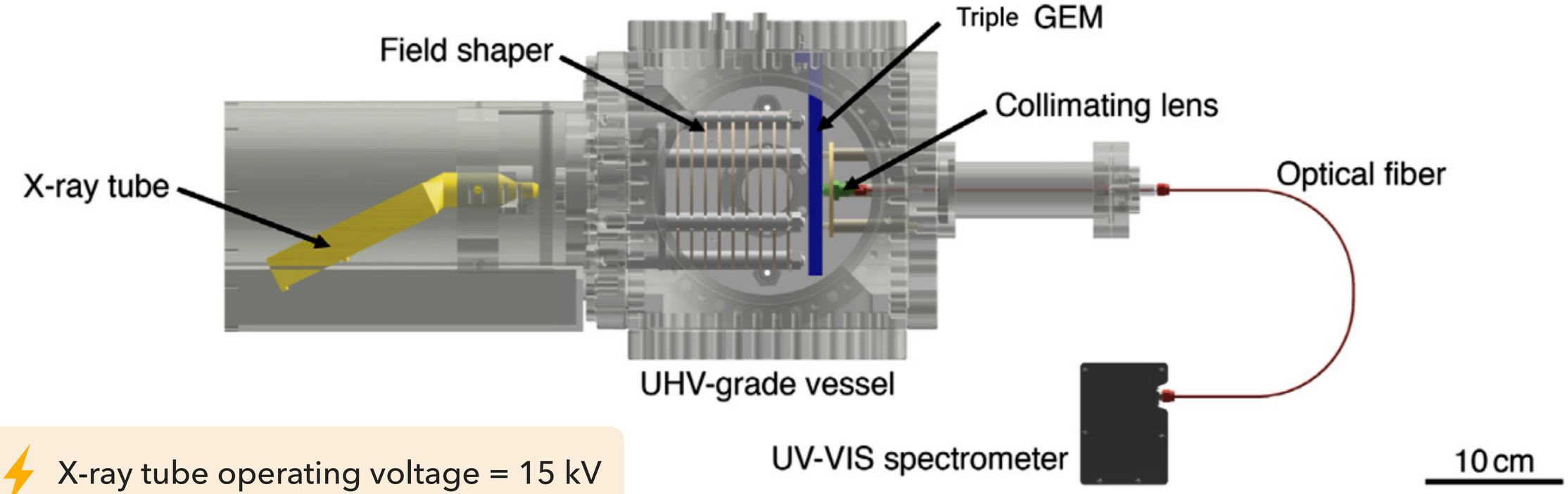
INCREASING HFO CONCENTRATION



INCREASING GEM VOLTAGE

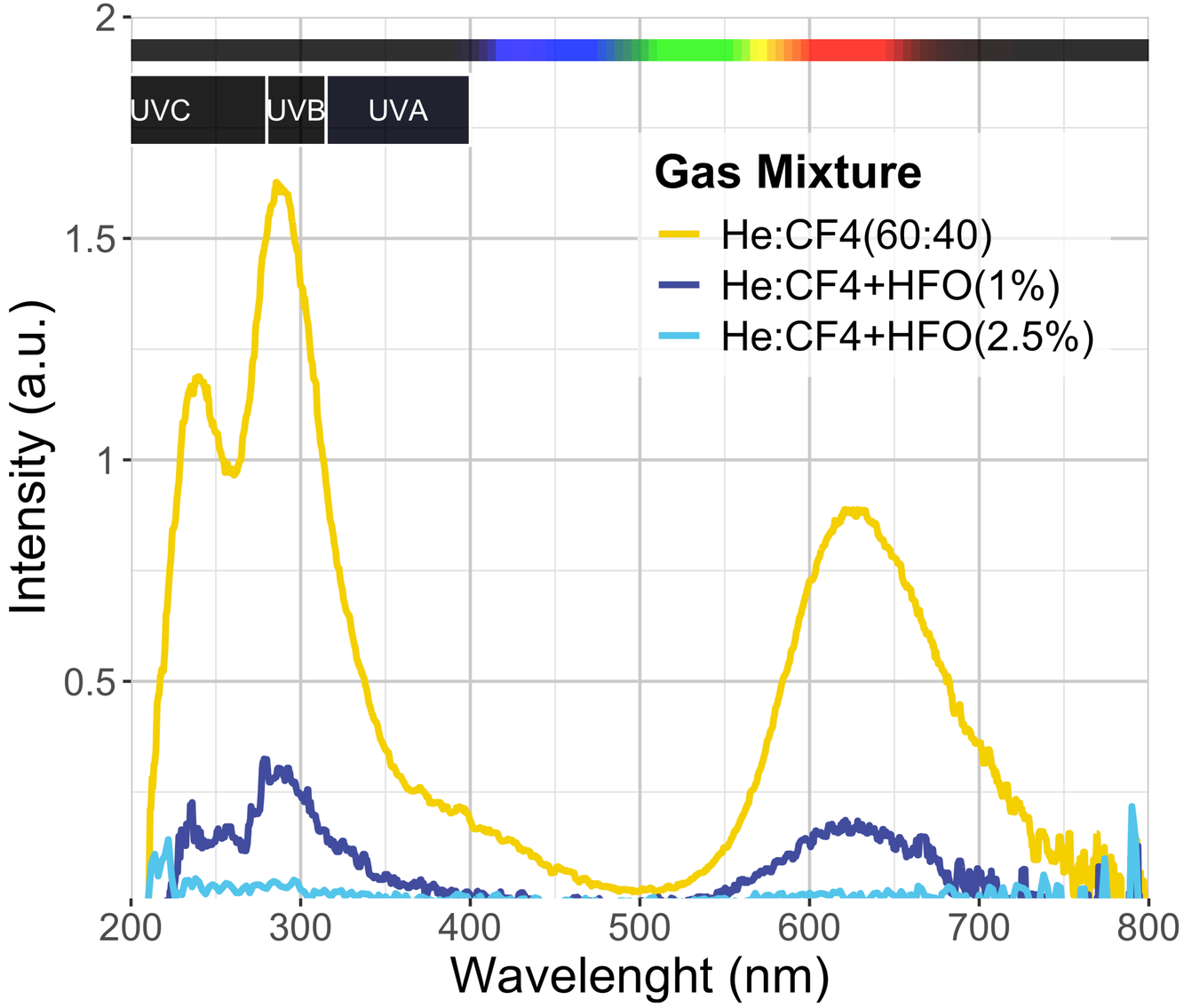
Data acquisition exposure time = 30 s

Current collected at the bottom of the third GEM monitored



X-ray tube operating voltage = 15 kV

Primary and secondary scintillation of CF4-based mixtures in low-pressure gaseous detectors
Brunbauer, F. M. et al. · Front. Detect. Sci. Technol. 3 (2025) 1561739 · doi:10.3389/fdest.2025.1561739



Two emission bands
Pure He:CF_4 emits in a UV band near 280 nm and a visible band near 620 nm.

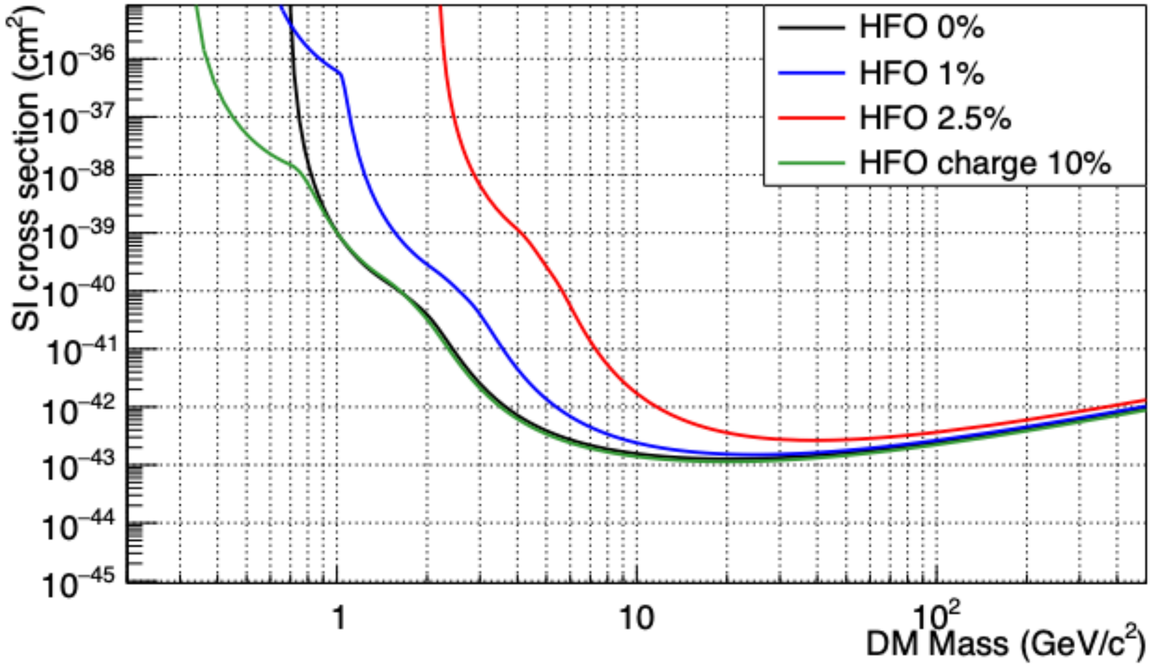
HFO strongly quenches
Just 1% HFO collapses the yield; at 2.5% the emission is almost entirely suppressed.

Both bands dim together
The UV:VIS ratio barely moves — HFO scales the whole spectrum down, it doesn't reshape it.

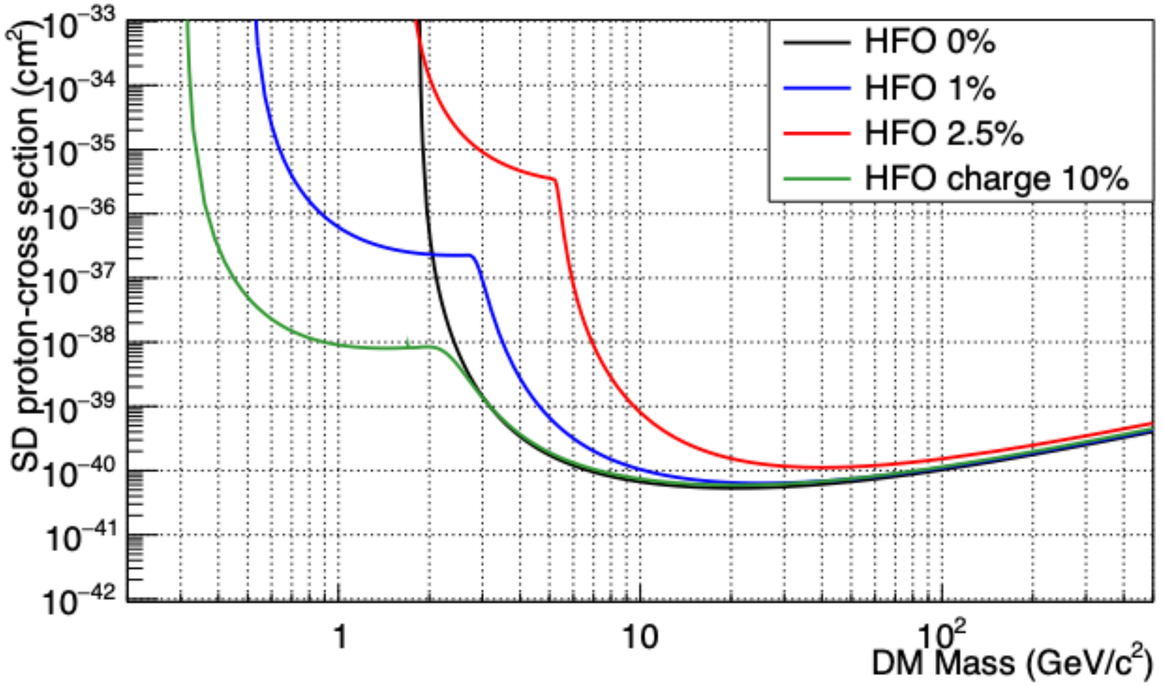
Edge bins are noisy
Spectrometer corrections amplify noise at the 200 / 800 nm extremes — read those with care.

Sets the optical ceiling
HFO-richer mixtures give too little light for reliable readout, so they were not pursued.

Bayesian analysis based on a 30 m^3 detector running for 3 years, yielding an **exposure of 153 kg·yr** and **10^4 background evt/year**



Mixtures with **low HFO content** (1%) provide the best balance of sensitivity, especially for **SD interactions**



The addition of **HFO** for a **pure charge readout** would allow for the best **low-mass DM sensitivity**

HFO-1234ze is a low-GWP route to hydrogen targets in CYGNO – it preserves charge gain but strongly quenches the light

01 **Stable gain up to 5% HFO.** Operation holds across the full range, needing higher GEM voltages.

02 **Light yield degrades rapidly.** Secondary scintillation is suppressed across UV and visible, and cannot be recovered within safe voltages.

03 **Optical sensitivity is threshold-limited.** Light quenching raises the energy threshold, so only percent-level HFO keeps a limited low-mass spin-dependent edge.

04 **Charge readout is the natural path.** Preserved gain makes low-GWP hydrogenated mixtures attractive for pure charge-readout schemes.